GDP and CO2 emissions: Record temperatures heighten urgency to break the link

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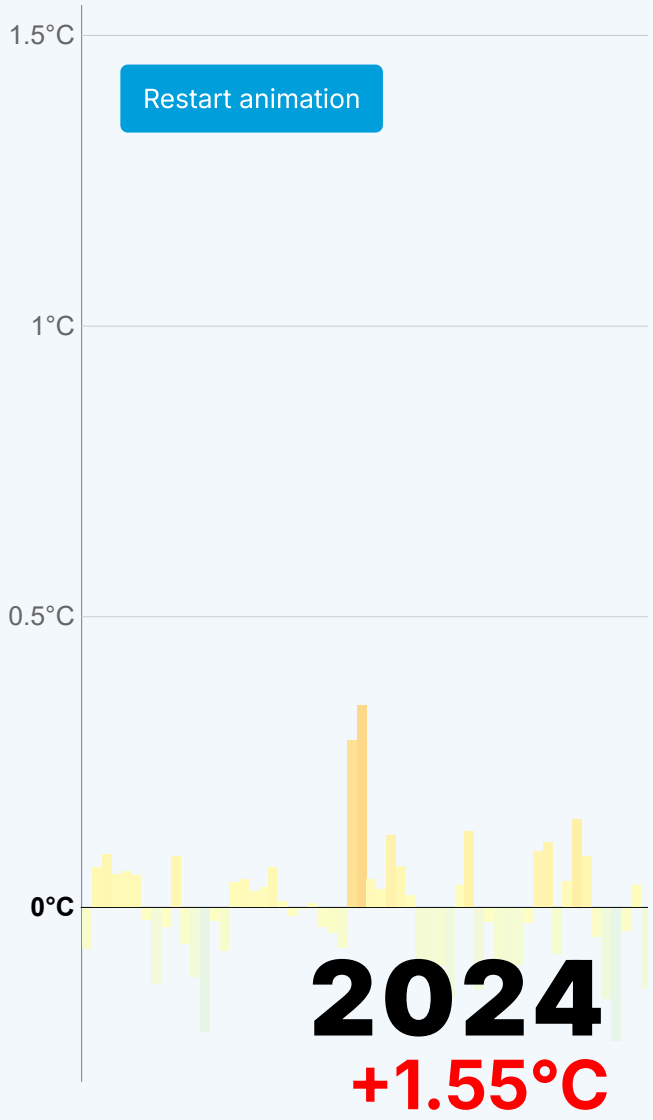
The link between economic growth and carbon emissions is weakening – but not fast enough to keep global heating well below 2°C, let alone limit it to 1.5°C.

Global heating pushes temperatures into danger zone

Annual global mean temperature anomalies relative to a pre-industrial (1850–1900) baseline, celcius, 1850–2024



Source: UN Trade and Development (UNCTAD), based on data from World Meteorological Organization (WMO) (https://wmo.int/publication-series/state-of-global-climate-2024). The annual average is calculated as the average from available data for each year.

Note: The WMO temperature assessment draws on six datasets: the European Center for Medium Range Weather Forecasts (ECMWF), Japan Meteorological Agency, NASA, the US National Oceanic and Atmospheric Administration (NOAA), the UK's Met Office in collaboration with the Climatic Research Unit at the University of East Anglia (HadCRUT), and Berkeley Earth.

The planet has crossed a dangerous threshold. The World Meteorological Organization says that **2024 was likely the first calendar year to have exceeded 1.5°C above pre-industrial levels** (https://wmo.int/news/media-centre/wmo-confirms-2024-warmest-year-record-about-155degc-above-pre-industrial-level).

The global mean near-surface temperature reached 1.55°C above the 1850–1900 average – a signal the world is getting perilously close to breaching the Paris Agreement's (https://unfccc.int/files/essential_background/convention/application/pdf/english_paris_agreement.p goal of keeping global heating well below 2°C and pursuing efforts to limit it to 1.5°C (https://www.un.org/en/climatechange/science/climate-issues/degrees-matter).

Carbon emissions from fossil fuels also hit a new high in 2024 – a record expected to be surpassed this year.

Turning the tide requires faster progress in breaking the link between economic growth and carbon emissions. Fossil fuels have powered prosperity for two centuries – lighting cities, fuelling transport and driving trade. That link is weakening, but not fast enough.

The GDP-emissions link is weakening – but too slowly

Data shows that some major economies have made encouraging progress.

Between 1990 and 2023, for example, **the European Union's net greenhouse gas emissions fell by about one third** while GDP grew by around two thirds, according to the European Environment Agency (https://www.eea.europa.eu/en/analysis/indicators/total-